

THRML Integration Architecture

● CORE COMPONENTS

CategoricalNode

└ Discrete state variables

SpinNode

└ Binary ± 1 variables

Block

└ Organization of related nodes

BlockSpec

└ Block structure specification

BlockGibbsSpec

└ Gibbs sampling configuration

● SAMPLING & FACTORS

CategoricalEBMFactor

└ Energy-based factor for categorical

CategoricalGibbsConditional

└ Categorical Gibbs sampler

FactorSamplingProgram

└ Factor-based sampling program

sample_states

└ Main THRML sampling function

SamplingSchedule

└ Warmup + sampling configuration

combines



● STATE MANAGEMENT

make_empty_block_state → block_state_to_global → from_global_state

THRML Methods Used: 14

Inference Accuracy (KL): 0.000751 nats

Status: ✓ All methods actively invoked